



CONCRETE WORKS COST ESTIMATOR PRO

Official Operation Manual

Version 1.0 | 2026

Desktop Application | Windows

dmvsoftware.gumroad.com

1. Introduction

Concrete Works Cost Estimator Pro is a professional desktop application designed to simplify and accelerate cost estimation for concrete construction projects.

The software combines automated geometric calculations, customisable productivity parameters, and structured cost management — helping estimators, contractors, engineers, and project managers create accurate, professional quotations with minimal manual effort.

The application supports both standard structural element calculations (Mode 1) and custom project-level estimating workflows (Mode 2), with multilingual interface support, multiple currencies, professional PDF reporting, and integrated estimate history management.

Who This Manual Is For

- Quantity surveyors and cost estimators
- Civil and structural engineers
- General contractors and subcontractors
- Project managers responsible for concrete works
- Any professional producing concrete construction quotations

Core Workflow Overview

Stage	Action	Output
DEFINE	Configure settings & enter project data	Productivity parameters, element type, dimensions or quantities
MEASURE	Run calculation	Volume, formwork area, reinforcement, labour hours, cost breakdown
VERIFY	Review, edit and generate PDF	Professional cost estimate / quotation ready for client delivery

2. Interface Overview & Initial Setup

Before creating an estimate, configure the application to match your working environment.

Navigation Bar

Control	Function
Mode 1 / Mode 2 buttons	Switch between Parametric and Manual calculation modes
Settings button	Open the Settings panel to configure productivity parameters
Currency selector (e.g. EUR)	Define the monetary unit used in estimates and PDF output
Language selector (e.g. EN)	Switch the interface language (8 languages supported)
Theme toggle	Switch between Light and Dark interface themes

Theme Selection

- Light Mode — recommended for bright office environments and printed reference
- Dark Mode — recommended for extended screen work

Language Selection

Click the language button to switch the interface language. Supported languages:

EN	ES	DE	FR	IT	EL	PL	PT
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Currency Selection

Use the currency selector to define the monetary unit. The selected currency appears throughout the interface and in generated PDF quotations.

3. Settings & Productivity Parameters

The Settings panel defines the default calculation parameters used by Mode 1.

NOTE

Settings values can be adjusted to reflect your company's historical project data, regional construction practices, or specific engineering specifications.

Adjustable Parameters

Parameter	Unit	Description
Formwork Ratio	m ² /m ³	Average formwork surface area per cubic metre of concrete for each element type
Labour Productivity	h/m ³	Estimated labour hours per cubic metre of concrete
Reinforcement Content	kg/m ³	Estimated reinforcement steel quantity per cubic metre of concrete

Supported Structural Element Types

Slab	Footing	Strip Foundation	Column	Beam	Concrete Wall
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4. Mode 1 — Parametric Calculation

Mode 1 is designed for fast, structured estimation of standard rectangular concrete elements. Enter project data and element dimensions — the application calculates all quantities and costs automatically.

Step 1 — Project Information

- Contractor name
- Client name
- Project name and description
- VAT number
- Country
- Estimate ID (auto-generated)

Step 2 — Structural Element & Concrete Class

Slab	Footing	Strip Foundation	Column	Beam	Concrete Wall
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Select the required concrete class (e.g. C25/30).

Step 3 — Geometry Input

Field	Unit	Notes
Length	m	Horizontal dimension of the element
Width	m	Perpendicular horizontal dimension
Thickness / Height	m	Vertical dimension (slab thickness or column/wall height)

The application automatically calculates:

- Concrete Volume (m³) = Length × Width × Thickness
- Formwork Area (m²) = Volume × Formwork Ratio (Settings)
- Reinforcement Weight (kg) = Volume × Reinforcement Content (Settings)
- Labour Hours (h) = Formwork Area × Labour Productivity (Settings)

Step 4 — Cost Input: Materials

Cost Field	Unit	Description
Concrete Unit Price	/ m³	Cost per cubic metre of concrete (excluding delivery)
Concrete Delivery Rate	/ m³	Delivery cost per cubic metre — added to concrete unit price
Reinforcement Unit Price	/ kg	Cost per kilogram of reinforcement steel
Formwork Unit Price	/ m²	Cost per square metre of formwork (supply and installation)

Step 5 — Cost Input: Labour & Equipment

Cost Field	Unit	Description
Labour Rate	/ h	Hourly labour cost (all-in rate including welfare and insurances)
Equipment & Services	total	Concrete pumping, vibration equipment, temporary services
Transport & Logistics	total	Site setup, equipment movement, additional logistics costs

Step 6 — Financial Parameters

Parameter	Description
Overhead (%)	Company overhead allocation applied to Direct Cost
Profit Margin (%)	Applied to (Direct Cost + Overhead)
VAT (%)	Value added tax applied to the Subtotal

Step 7 — Quantity Multiplier

Use the Quantity field to estimate multiple identical elements in a single calculation.

EX	4 identical columns → Quantity = 4. All totals are multiplied automatically.
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Step 8 — Generate Estimate

Click **Calculate** to generate the full cost breakdown. Click **Review** to open the quotation preview.

5. Mode 2 — Manual / Custom Calculation

Mode 2 is designed for flexible estimating when standard geometric calculation is not applicable. Quantities are entered directly from engineering drawings or professional takeoff calculations.

When to Use Mode 2

- Non-standard or irregular structures
- Circular, polygonal, or curved elements
- Complex architectural concrete
- Staircases and ramps
- Complete building-level concrete takeoffs
- Projects where all quantities are already calculated externally

Manual Quantity Input

Field	Unit	Source
Total Concrete Volume	m ³	Engineering drawings / structural calculations
Total Formwork Area	m ²	Detailed formwork takeoff
Total Reinforcement Weight	kg	Bar bending schedule or BBS
Total Labour Hours	h	Resource schedule or productivity estimate

Smart Transfer from Mode 1

When switching from Mode 1 to Mode 2, the application automatically transfers calculated quantities (Volume, Formwork Area, Reinforcement Weight, Labour Hours) into the Mode 2 input fields. All cost rates and financial parameters are preserved.

TIP

Use Mode 1 first to set up all cost rates and project data, then switch to Mode 2 to override quantities with values from your detailed takeoff.

6. History & Data Management

The application includes an integrated estimate history system that automatically saves each completed calculation for future reference and reuse.

Automatic Saving

Field	Content
Date & Time	Timestamp of the calculation
Estimate ID	Unique estimate reference number
Client	Client name as entered
Project	Project description
Element Type	Structural element type (Mode 1) or custom (Mode 2)

Total Cost	Final quotation value including VAT
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Search

Use the search field to locate previous estimates by client name, project name, or Estimate ID. Results update as you type.

Recall Previous Estimates

Double-click any history entry to automatically restore the complete saved estimate data. All fields are repopulated — allowing fast quotation revisions, template creation, or repeated calculations with updated prices.

Clipboard Export

Select a history entry to copy summary estimate information for use in external applications such as spreadsheets or email.

7. PDF Reporting & Email Export

After completing a calculation, generate a professional PDF quotation document ready for client delivery.

Review & Editing

Click the **Review** button to open the quotation preview window. You can edit:

- Scope of Works — describe the work included in the quotation
- Notes — add payment terms, validity period, exclusions, or other conditionsClick **Apply and**

Save to confirm edits and generate the final PDF.

PDF Content

#	Section
1.	Project header (Contractor, Client, Project, Date, Estimate ID)
2.	Project quantities breakdown
3.	Materials cost section
4.	Labour cost section
5.	Equipment & Logistics section
6.	Cost summary (Direct Cost → VAT → Total Cost)
7.	Scope of Works and Notes
8.	Signature section and QR verification code

QR Code Verification

Each PDF quotation contains a QR code encoding the estimate's unique verification hash, allowing independent verification of the document's authenticity.

Email Integration

Click the **Send** button to open your default desktop email application with the PDF automatically attached and the Estimate ID pre-filled as the email subject.

NOTE

The Send function uses the operating system's default email application (e.g. Outlook, Thunderbird). Ensure a default email client is configured on your system.

8. Best Practice Recommendations

Settings & Parameters

- Update productivity parameters regularly using data from completed projects.
- Verify Formwork Ratio values against your typical construction methods.
- Adjust Reinforcement Content to reflect project-specific engineering specifications.
- Cross-reference Labour Productivity rates with your resource planning data.

Estimating Workflow

- Use Mode 1 for standard repetitive structural elements (slabs, footings, columns, beams, walls).
- Use Mode 2 for complex shapes, non-standard structures, or full project-level takeoffs.
- Use the Smart Transfer feature when switching from Mode 1 to Mode 2.
- Always use the Quantity multiplier for repeated identical elements rather than recalculating.

Cost Input

- Verify all unit cost inputs before generating the final quotation.
- Include all applicable costs in Equipment & Services and Transport & Logistics fields.
- Review the financial parameters (Overhead, Margin, VAT) before each new project.

PDF Quotations

- Always use the Review function to check Scope of Works and Notes before generating the PDF.
- Ensure the Estimate ID is meaningful and traceable to your internal project reference system.
- Send quotations from a configured professional email account.

History Management

- Use descriptive Client and Project names to make history search effective.
- Restore previous estimates as templates for similar new projects.
- Periodically review saved estimates to monitor unit cost trends over time.

TIP

Concrete Works Cost Estimator Pro is designed to provide a structured, reliable workflow from initial quantity input to professional client-ready documentation. The more accurately the Settings parameters reflect your real project data, the more reliable the resulting estimates.

9. Quick Reference — Calculation Logic

The following table summarises the automatic calculations performed by the application in Mode 1 (Parametric). All formulas reflect the actual computation logic in the software.

Quantity	Formula	Notes
Concrete Volume	$L \times W \times T$	Direct geometric calculation (Length $\times$ Width $\times$ Thickness)
Formwork Area	Volume $\times$ Formwork Ratio	Ratio from Settings (m ² /m ³)
Reinforcement Weight	Volume $\times$ Reinforcement Content	Content from Settings (kg/m ³)
Labour Hours	Formwork Area $\times$ Labour Productivity	Productivity from Settings (h/m ³)
Concrete Cost	Volume $\times$ Concrete Unit Price	Excluding delivery
Delivery Cost	Volume $\times$ Delivery Rate	Added to concrete cost
Reinforcement Cost	Reinf. Weight $\times$ Unit Price	kg $\times$ price per kg
Labour Cost	Labour Hours $\times$ Labour Rate	h $\times$ hourly rate
Formwork Cost	Formwork Area $\times$ Formwork Unit Price	m ² $\times$ price per m ²
Equipment & Logistics	Equipment + Transport inputs	Direct lump sum inputs
Direct Cost	Concrete + Delivery + Reinforcement + Labour + Formwork + Equipment & Logistics	Pre-overhead total
Overhead Cost	Direct Cost $\times$ Overhead %	Company overhead allocation
Margin Cost	(Direct Cost + Overhead) $\times$ Margin %	Applied after overhead addition
Subtotal	Direct Cost + Overhead + Margin	Pre-VAT total
VAT Amount	Subtotal $\times$ VAT %	Applied to full subtotal
Total Cost	Subtotal + VAT Amount	Final quotation value

Quantity Multiplier

When Quantity > 1, all calculated values (volumes, areas, weights, hours, and all costs) are multiplied by the Quantity value before display and before being saved to history.

Mode 2 — Manual Input

In Mode 2, Concrete Volume, Formwork Area, Reinforcement Weight, and Labour Hours are entered directly by the user. All cost calculations from Labour Cost onward follow the same formulas as Mode 1.